Cole Kennedy

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Education

University of South Carolina Expected Dec. 2027
***Bachelor of Science in Computer Science**; GPA: 3.93* *Columbia, SC*

- Academic Scholar - Distinction Award, Capstone Scholar, President's Honor List 3x
- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Software Engineering, Computer Architecture, Advanced Programming Techniques (C++), Discrete Mathematics, UNIX/Linux Fundamentals

Experience

Software Engineer Intern Jun. 2026 – Present
Evolv Technology *Waltham, MA*

- Owned the Collimator Alignment feature end-to-end for Evolv's eXpedite system, rebuilding a legacy laptop tool into an integrated React/TypeScript interface for technician alignment of live X-ray detector hardware
- Engineered a real-time gRPC-streaming and Electron IPC pipeline ingesting dual-energy X-ray frames, computing per-board gain across 8 boards × 64 columns against a simulated hardware backend
- Designed custom SVG visualizations and a multi-phase alignment workflow rendering live detector traces, limit bands, and pass/fail feedback during real-time hardware calibration
- Modeled detector gain and mapped statistical limits into native count space using production firmware values
- Integrated the UI with embedded-Linux C++ services via gRPC/Protobuf, orchestrating diagnostics setup mode through database-flag writes, reboot control, and system-status-driven navigation

Software Engineer Intern Oct. 2025 – May 2026
Knead *Remote*

- Built a Python code generation pipeline to translate user-defined trading strategies into executable backtesting code, enabling flexible composition of event-driven strategies from reusable strategy components
- Designed an authentication architecture using AWS Cognito and implemented JWT-based authorization backed by Supabase, securing backend APIs and enforcing user-level data access across services
- Developed a Python simulation and validation framework with asynchronous backtesting, reducing end-to-end strategy evaluation time by ~40% through parallel execution
- Refactored live strategy execution services into a stateless, multi-tenant backend with persistent executor state, per-user isolation, and recovery of active strategies after server restarts

Leadership & Teaching Experience

Instructional Assistant Aug. 2025 – Present
University of South Carolina *Columbia, SC*

- Led weekly peer instruction sessions on advanced C++ topics including object-oriented design, pointers, templates, and memory management, with emphasis on correctness, performance, and debugging
- Designed and taught interactive coding assignments that integrate data structures, algorithmic analysis, and debugging strategies to help students apply theory to practical programming problems

Projects

Database Engine | ***C++, STL, React, TypeScript*** *2026*

- Built a disk-backed key-value database engine in C++ with page-based storage, custom record serialization, overflow page handling, persistent B+ tree indexing, buffer pool caching, and crash-aware file validation
- Developed a REST API and React dashboard for record management, B+ tree visualization, and database metrics

Asynchronous Job Processing Service | ***Python, FastAPI, Redis, PostgreSQL, Docker*** *2025*

- Built an async job service using Redis queues and workers to decouple API handling from background execution
- Implemented PostgreSQL job state management with retries, idempotency, and fault-tolerant lifecycle handling

Technical Skills

Languages: Python, TypeScript, JavaScript, C++, Java, SQL, HTML, CSS
Frameworks/Libraries: React, Electron, FastAPI, Flask, Tailwind CSS
Technologies: gRPC, Protobuf, PostgreSQL, Supabase, Redis, Docker, Git, Linux, AWS Cognito